

CLOUD-2 — #521: the batched journal is not backward-readable — decide the rollback story, honestly

Batch of 2026-08-26 · merge order 2 of 5 (before CLOUD-3 — you share the activity/journal subsystem; your format decision lands first).

```
task_id: gv-521-cloud-2
issue: 521
branch: cloud/521-journal-rollback
base: main          # 682f3061 or later
adr_number: 0085    # ASSIGNED — whether you amend ADR 0080 or write fresh, the
                    # number is reserved either way
sign_commits_as:
  name: Claude_Max
  email: 262510778+tom2025b@users.noreply.github.com
  # per-commit, ALWAYS: git -c user.name=Claude_Max -c
  user.email=262510778+tom2025b@users.noreply.github.com commit ...
allowed_paths:
  - crates/git-vista-server/src/journal.rs
  - crates/git-vista-core/src/activity.rs
  - crates/git-vista-server/src/**          # reader/writer seams as needed
  - docs/adr/
forbidden_paths:
  - crates/git-vista/src/**
  - ci/browser/**
deliverables:
  - branch pushed, PR opened with "Closes #521"
  - ADR 0085 (or an explicit amendment to 0080's consequences carrying the 0085
    number in the index)
```

Environment truths — read before your first test run

- **~320 server tests fail in your container on unmodified main** (missing Landlock; sandbox tier refuses). Baseline on unmodified main FIRST; only the diff is yours; both counts in the PR body.
- **cargo build -p git-vista-server --bin gv-sandbox before any server test run** — the helper is not built by `--bin git-vista-server` selection and hundreds of tests exec it at runtime.
- **Browser leg unrun** — say so in the PR body. Owner runs it before merge.

Truth-checked state (verified against main 682f3061, 2026-08-26)

`RefsAtEvent` is an enum in `crates/git-vista-core/src/activity.rs:175`, with an `InBatch { batch }` variant resolved at `activity.rs:274-276`. The #485 batching (ADR 0080, PR

#498) made a 500-ref fetch journal one full snapshot plus N-1 `in_batch` referrers. #519 (merged 2026-08-25) put the writing pid into batch ids — read its commit before touching id semantics.

The defect, plainly

A pre-#485 binary's `RefsAtEvent` has no catch-all variant, and its `read_all` skips unparseable lines. **After a rollback, every `{"status": "in_batch"}` referrer is an unknown variant and vanishes:** a 100-ref fetch renders as one event until re-upgrade. Bytes on disk are intact — rendering loss, not data loss — but it is a persisted-format break that ADR 0080 never weighed.

The job — a decision first, then the smallest honest mechanism

ADR 0080 skipped an alternative: a versioned batch envelope in the same JSONL stream, where an old reader drops ONE unsupported record instead of N-1. Your job:

1. **Analyze both options against the actual reader code** (old readers are shipped binaries — nothing you write fixes THEM; be precise about what each option buys for the past vs the future): - (a) accept the rollback cost and document it in ADR 0080's consequences, verbatim about the N-1 loss; - (b) additionally stamp a **journal format version** on new writes with a tolerant reader, so the NEXT format change has an answer at read-time instead of a guess (this is the same lesson #509 is teaching the durable operation store today, and the M5-family reviews flagged both).
2. **Recommend and implement.** The house prior is (a)+(b): document the cost that is already sunk, and version what is still cheap to version. If your analysis lands elsewhere, the ADR argues it — a well-argued (a) alone beats a mechanical (b) nobody can state the benefit of.
3. Whatever ships: the version stamp (if any) must not break the CURRENT reader on files written by main's binary — mixed-line files are the normal case, not an edge. Pin that with a fixture holding pre-batch, batch, and (if introduced) stamped lines in one file.
4. **Mutation-prove two different ways** any invariant you pin (e.g. drop the tolerant-read arm; then mis-stamp the version on write), red at different assertions, byte-identical restore verified.

Acceptance

1. ADR 0085 records the decision, the rejected alternative, and the honest statement of what rollback still costs after your change.
2. Mixed-line fixture test exists and passes; mutation evidence in the PR body (two red assertion lines, verbatim).
3. `cargo fmt --all · clippy --all-targets -- -D warnings · server suite` zero new failures vs your recorded baseline.
4. PR body: baseline counts, browser-leg-unrun line, ADR summary, your session tag.

Written by fable · 2026-08-26 · truth-checked against 682f3061 the same morning.